

Chapter 159

Resolve FX Key

DaVinci Resolve includes several Resolve FX dedicated to compositing and keying tasks that are applied directly on the Timeline.

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3D Keyer

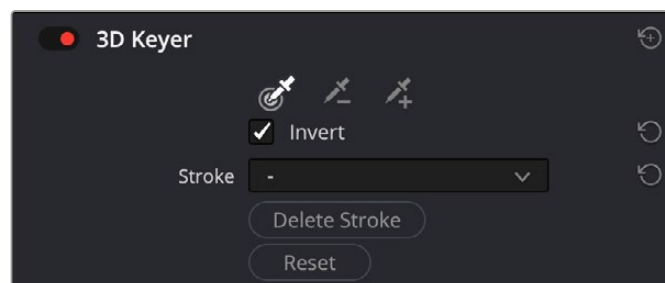
Based on the 3D Qualifier in the Color page, the 3D Keyer Resolve FX offers a fast, simple way of pulling a key to isolate a range of color in the image by dragging your mouse pointer over the parts of the image you want to key. Each time you drag over the image to add or subtract from the cloud of values, you're carving out of a three-dimensional representation of all available colors; this is viewable in the tools Key Map box and is what gives the 3D Keyer its name.

TIP: If you're looking to isolate a range of luma values in the image, the HSL or Luma keyer will give you more control.

The 3D Keyer is a good one to start with if you're trying to key a blue screen or green screen background. Its interface of dragging the mouse pointer over the screen color you want to remove, with live feedback, gives precise control. The 3D Keyer's greatest strength is the speed with which you can sample areas of the picture to extract from the final key. However, this can also be a weakness if your initial samples aren't giving you satisfactory results because, unlike the Delta Keyer in the Fusion page, there aren't many ways to fine-tune the key as it's being generated. On the other hand, for well-lit subjects you want to key, two or three samples is all you need. Small numbers of long strokes will generally give better results in consistently selecting a range of similar colors, like a broad brush. Conversely, five to ten short strokes act like a fine brush and will only select a very narrow range of color that you've specified.

Selection Range Controls

The Selection Range buttons in the Inspector let you define a key by sampling pixels in the Viewer with the mouse pointer.



The 3D Keyer Selection Range controls

- **Picker:** Chooses the initial color to qualify. Longer strokes will tend to give you a better key. In the Edit page, the Timeline Viewer Overlay must be set to Open FX.
- **Picker -:** Lets you choose an additional color to exclude from the current qualification; also available by holding down the Option key.
- **Picker +:** Chooses an additional color region to add to the current qualification.
- **Invert:** Inverts the current key; qualified areas are then unqualified and vice versa.
- **Stroke:** Displays the list of picked (+) and unpicked (-) colors, along with their respective RGB values.
- **Delete Stroke:** Deletes the current stroke selected in the Stroke drop-down menu.
- **Reset:** Deletes all strokes and resets the 3D Keyer to its default state.

Behavior Options

These controls adjust the fundamental parameters and modes of the 3D Keyer.

- **Colorspace:** This drop-down menu lets you choose whether you're sampling colors in YUV, HSL, HSP, or LAB colorspaces. YUV is the default, but if you find you're not getting satisfactory results, you can choose another colorspace from the drop-down menu to see if that works any better. These have various theoretical differences but the best choice always depends on the task.
- **Soft:** Provides a softer edge to the key that is more forgiving of Chroma and Luma adjustments. The more gentle drop-off is suitable for situations like subtle light changes across a face. Use the Shadow/Midtone/Highlight Matte Finesse controls in conjunction with this mode to fine tune the result.
- **Flat:** The default mode. Each color selected is 100% keyed, and adjustments are made for small color variations. This is the recommended mode for chroma keying a green or blue screen. Additionally, turning up the Pre-Filter setting in the Matte Finesse controls can make for a smoother, flatter key.
- **Tight:** Only keys the exact color range picked and does not apply any softening based on color ranges. Single pixel sharpness levels are expected. This can be used for difficult keying jobs, requiring adjusting the filter and softness manually in the Matte Finesse controls.
- **Luma:** Functions similar to the Tight setting but ignores all chroma data, letting you key by brightness level and not color.
- **Despill:** If you're using the 3D qualifier to pull a blue or green screen key to create transparency, this slider lets you adjust color correction that eliminates blue or green spill from the image, while retaining the image's original color.

Usage Options



The Key Map, shown in the upper left, showing a qualifier that attempts to select the sand in the background (blue lines), while deselecting the pirate's hats (red lines)

These settings control the user feedback tools from the 3D Keyer in the Viewer. In the Edit page, the Timeline Viewer Overlay must be set to Open FX.

- **Show Paths:** A checkbox that lets you turn on and off the visibility of the lines you're drawing to sample the image. Turning lines off does not affect the key.

- **Smart Show Paths:** With this checkbox enabled, the lines will only be visible as they are being drawn, and disappear when the mouse button is released.
- **Auto-B/W Highlight:** With this checkbox enabled, the Viewer automatically switches to Highlight B/W mode while drawing a line, to better show you the resulting key in real time. When the mouse button is released, the Viewer switches back to its original viewing mode.
- **Show Key Map:** Check this box to show the Key Map in the Viewer. The Key Map is composed of the Color Space box, and the Brightness Range. The Color Space box is laid out roughly in the same manner as the Color Wheels, with primary colors laid out around its edge. The exact colors and layout is determined by the Color Space that you choose in the 3D Keyer. The Brightness Range is represented as a bar underneath the Color Space, with left being black and right being white. Inside the Color Space and Brightness Range lie the individual colors that you chose with the pickers in the Strokes List.
- **Key Map Zoom:** Controls the zoom level of the Key Map, so you can see finer detail, if necessary.

Key Adjustments

This set of tools allows you to manually adjust the generation of the key from the sampled colors. Feedback from these tools can be seen in the Key Map, if enabled.

- **Chroma Tolerance:** Click and drag left and right to expand or contract the range of colors selected by the key.
- **Chroma Softness:** Click and drag left and right to change the sensitivity to similar colors selected by the key. This determines whether the key is a hard cutoff or a soft selection of similar colors.
- **Adaptive Chroma Softness:** Checking this box allows the keyer to operate consistently in both highly saturated and desaturated regions of the image at the same time. The majority of the time you will want to leave this on. The exception is if you are having difficulty in manually adjusting chroma softness in Soft or Flat mode, then turning Adaptive Chroma Softness off will give you more range to work with. Adaptive Chroma Softness is automatically disabled in Tight and Luma modes as it is contrary to those mode's functionality.
- **Chroma Tilt:** Moves your selected color region within the color space, changing its chroma based on its vertical position.
- **Chroma Shift:** Moves your selected color region within the color space, changing its chroma based on its horizontal position.
- **Chroma Rotate:** Moves your selected color region within the color space, changing its tint and temperature based on its rotation around the central origin point.
- **Luma Low:** Expands or contracts the dark areas of the key in the brightness range.
- **Luma High:** Expands or contracts the bright areas of the key in the brightness range.
- **Low Softness:** Controls how well defined the low end of the brightness range is. This determines whether the key is a hard cutoff or a soft selection of similar brightness levels.
- **High Softness:** Controls how well defined the high end of the brightness range is. This determines whether the key is a hard cutoff or a soft selection of similar brightness levels.

TIP: All of the parameters above are key-framable, allowing you to animate the key over time. For example, you may have qualified a shirt where the actor walks through different colored light in a scene. By key framing the parameters above, you can hold the same key on the shirt as it changes colors over time.

Matte Finesse

You may find that sampling the screen color is not enough to overcome some problems. Issues such as chattery edges, holes, or noise can sometimes be fixed using the Matte Finesse controls. These controls filter the output of the Keyer and are adjustments that are made to the matte itself, modifying the key using the frame's content.

- **Pre Filter:** This slider attempts to clean up the image before colors are sampled. This adjustment can be useful when you have footage containing MPEG blocking artifacts.
- **Clean Black:** Clean Black is a specialized operation that eliminates noise seen as white speckling in the black area of a key. Raising Clean Black lets you “fill holes” in the background portion of a key and erode translucent edges.
- **Clean White:** Clean White is another specialized operation that eliminates noise (seen as black speckling when viewing a high-contrast highlight) in the white portion of a key that includes areas of the image you're isolating and expands the key by making light parts of a key lighter the higher you raise this parameter, pushing light gray areas of the key toward white. The practical result is that raising Clean White lets you “fill holes” in the foreground portion of a key and grow translucent edges.
- **Black Clip:** Raising Black Clip applies a “lift” adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed toward black. The range is 0 to 100, with 0 being the default setting.
- **White Clip:** Lowering White Clip applies a “gain” adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed toward white. The range is 0 to 100, with 100 being the default setting.
- **Blur Radius:** In small amounts, blurring a key does well to take the edge off problem edges. However, blurring a key can also feather the edges of a key past the border of the subject you're keying, with the result being a visible “halo” around your subject, depending on the adjustment you're making. The range is 0 to 2000, with 0 being the default. With such a large maximum blur radius, combined with the capabilities that the In/Out Ratio provides in customizing the direction of spread, you can turn some pretty precarious mattes into surprisingly smooth and useful results.
- **In/Out Ratio:** Using In/Out Ratio can help eliminate fringing. Raising In/Out Ratio will fill in small black holes in the matte, while lowering In/Out Ratio below 0 will eliminate speckling by pushing small white bits of the matte toward black.
- **Morph Operation:** You can choose Shrink or Grow to dilate or erode the edges of the matte, or you can choose Opening or Closing to plug or expand holes to clean up a ragged matte.
- **Morph Radius:** Adjusts how much to shrink, grow, open, or close the key.
- **Denoise:** Provides a distinct way to post-process extracted keys to selectively reduce the noise in a key, getting rid of stray areas of qualification and softly filling holes in a matte.
- **Shadow:** Adjusts key strength based on the darker parts of the original image.
- **Midtone:** Adjusts key strength based on the midtones of the original image.
- **Highlight:** Adjusts key strength based on the brighter parts of the original image.
- **Post Filter:** Performs a final clean-up of the key, using the original image for reference; useful for bringing back some fine detail in sharp edges or hair.

Garbage Matte

These controls let you crop out other unwanted things in the frame, such as lighting fixtures, boom microphones, tracking patches, etc.

- **Matte Shape:** You can choose either a Rectangle or Ellipse with which to crop around the subject you're keying. Choosing None turns off Garbage Matte. When a garbage matte is enabled, the Open FX Overlay mode of the Viewer shows an on-screen control for adjusting the shape and position of the garbage matte.
- **Edge Softness:** Lets you soften the edges. Defaults to 50; the range is 0 (no softening) to 100 (maximum softening).
- **Invert:** Inverts the garbage matte.
- **Center X and Y:** Lets you adjust and keyframe the position of a garbage matte.
- **Bounding Width and Height:** Lets you adjust the width and height of the garbage matte.
- **Rotation:** Lets you adjust the angle of the garbage matte.

Output

The Output controls let you choose how the image is output, composited, and displayed in the Viewer.

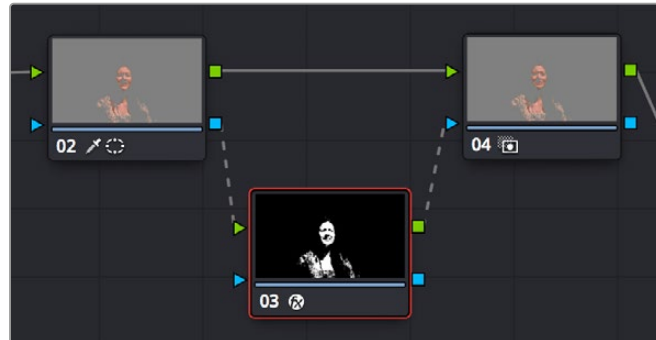
- **Output:** This menu offers three options for which channels are output from the effect.
 - Alpha Highlight:** Displays the transparent part of an Alpha channel as gray and the solid part of an alpha channel in full color.
 - Alpha Highlight B/W:** Displays the black and white Alpha channel, which is often helpful when sampling additional areas to key.
 - Final Composite:** The keyed image is displayed composited over any video track below it.
- **Use Alpha:** A common control in many Resolve FX and 3rd party Open FX plugins; this checkbox determines if the Alpha channel is used when compositing over another video track. It can be used instead of choosing RGB Only (Blank Alpha) from the Output menu.

To use the 3D Keyer in the Edit page:

- 1 Apply the 3D Keyer to a green screen or blue screen clip that's on a higher track than the background.
- 2 In the Inspector, enable the Show Paths checkbox to view the sampled areas once you begin.
- 3 From the Viewer Overlay menu in the lower-left corner of the Timeline Viewer, choose Open FX Overlays.
- 4 In the Viewer, click and drag to draw a sampling stroke across the screen color you want to remove.
- 5 The transparency immediately takes effect, showing the keyed subject against whatever clip appears on the video track underneath it in the Timeline as a composite. To see the matte you created for further adjustment, choose Alpha Highlight or Alpha Highlight B/W from the Output drop-down menu.
- 6 If necessary, you can add or subtract areas from the screen color selection using the add/subtract Color range controls. Alternatively, with the default Sample eyedropper selected, you can hold down Shift to add to the screen color selection or hold down the Option key to remove from the screen color selection.

Alpha Matte Shrink and Grow (Studio Version Only)

This filter lets you refine the edges of alpha and key mattes in a variety of ways, shrinking and growing the edges, and opening and closing holes that appear within a matte. In the Color page, this filter can be applied to Qualifier or Window keys that have been routed to the RGB input of a corrector node, for adjustment and refinement, before connecting the result to the key input of a subsequent node for use isolating an adjustment.



Applying the Alpha Matte Shrink and Grow filter to a key matte that's fed to the RGB input of a separate node

- **Morph Operation:** A drop-down menu lets you choose how you want to modify the Alpha channel/key. You can choose Shrink or Grow to dilate or erode the edges of the matte with great accuracy. Or, you can choose Opening or Closing to plug or expand holes within the key to clean up a ragged matte.
- **Shape:** A drop-down menu lets you choose how corners and angles in the edges of the key are handled when you grow or shrink it. Circle is the default, and results in even expansion around the surface of the key that eventually averages all angles into a circular shape if you use extreme Radius settings. Square averages all angles into a more rectangular shape if you use extreme Radius settings. Diamond averages all angles into a diamond shape if you use extreme Radius settings.
- **Radius:** A slider adjusts how much to Shrink, Grow, Open, or Close the key.
- **Process:** Lets you select specific channels to apply the Morph Operation to. Options are RGB (Blank Alpha), Alpha (RGB Unaffected), and RGB + Alpha.

Preview Alpha: Checking this box lets you quickly see the Alpha channel that will be output from the node.

- **Manual Iteration Control:** Checking this box gives you direct control on how the operators are applied.

Operator Radius: A slider that controls the size of the operator.

Repeat Operation: A slider alters the effect of the Operator Radius setting to create more extreme adjustments.

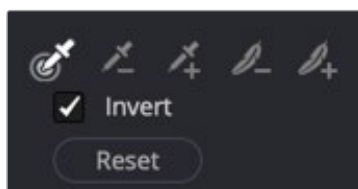
HSL Keyer

Based on the HSL Qualifier in the Color page, the HSL Keyer Resolve FX is a general-purpose keyer that uses three color components, hue, saturation, and luma, to define a key. The HSL Keyer also gives you the option of disabling color components that you don't want to contribute to the final key. You can pull a saturation-only key, or a hue-only key, for instances where choosing only individual components gives a better result.

The most straightforward way to use any keyer is to sample the image in the Viewer with no changes to the settings. When you sample one or more pixels, the hue, saturation, and luma values are analyzed and set different ranges. There are several eyedropper controls available to sample color and soften the selection.

Selection Range Controls

The Selection Range buttons in the Inspector let you define a key by sampling and softening pixels in the Viewer with the mouse pointer.

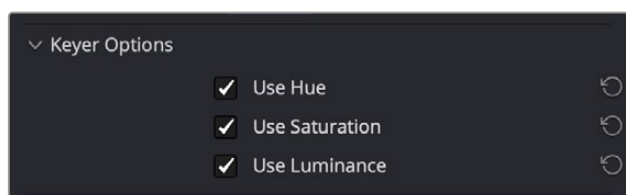


HSL Keyer Color Range controls

- **Sample Eyedropper:** This tool selects the colors to use as the range of qualification by clicking and dragging over the image in the Viewer. Clicking it again clears the previous selection. Generally, you will get better results using long drags instead of short ones.
- **Add/Subtract Color Range:** These two controls let you add areas of the image to, or subtract them from, the currently selected inner range of values that define the core of the key. Like the other sampling controls, you can drag over a range of color.
- **Add/Subtract Softness:** These two controls let you redefine the softness that transitions from the inner range of the key, falling off towards the outer edge of the key. Like the other sampling controls, you can drag over a range of color.
- **Invert:** Inverts the sampled color to become opaque, and areas not sampled become transparent.
- **Reset:** Resets all the sampled colors while retaining the general control settings.

Keyer Options

One of the HSL Keyer's strengths is that you can enable or disable each of the three HSL components using the Hue, Sat, and Lum checkboxes. This lets you ignore specific color components while focusing on others that may be more important. For example, if you're trying to key the saturated parts of an image, regardless of the hue or brightness, you can turn off Hue and Lum so that only Sat is used to sample it.



HSL Keyer options

Matte Finesse

You may find that sampling the screen color is not enough to overcome some problems. Issues such as chattery edges, holes, or noise can sometimes be fixed using the Matte Finesse controls. These controls filter the output of the Keyer and are adjustments that are made to the matte itself, modifying the key using the frame's content.

- **Pre Filter:** This slider attempts to clean up the image before colors are sampled. This adjustment can be useful when you have footage containing MPEG blocking artifacts.
- **Clean Black:** Clean Black is a specialized operation that eliminates noise seen as white speckling in the black area of a key. Raising Clean Black lets you “fill holes” in the background portion of a key and erode translucent edges.
- **Clean White:** Clean White is another specialized operation that eliminates noise (seen as black speckling when viewing a high-contrast highlight) in the white portion of a key that includes areas of the image you're isolating and expands the key by making light parts of a key lighter the higher you raise this parameter, pushing light gray areas of the key toward white. The practical result is that raising Clean White lets you “fill holes” in the foreground portion of a key and grow translucent edges.
- **Black Clip:** Raising Black Clip applies a “lift” adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed toward black. The range is 0 to 100, with 0 being the default setting.
- **White Clip:** Lowering White Clip applies a “gain” adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed toward white. The range is 0 to 100, with 100 being the default setting.
- **Blur Radius:** In small amounts, blurring a key does well to take the edge off problem edges. However, blurring a key can also feather the edges of a key past the border of the subject you're keying, with the result being a visible “halo” around your subject, depending on the adjustment you're making. The range is 0 to 2000, with 0 being the default. With such a large maximum blur radius, combined with the capabilities that the In/Out Ratio provides in customizing the direction of spread, you can turn some pretty precarious mattes into surprisingly smooth and useful results.
- **In/Out Ratio:** Using In/Out Ratio can help eliminate fringing. Raising In/Out Ratio will fill in small black holes in the matte, while lowering In/Out Ratio below 0 will eliminate speckling by pushing small white bits of the matte toward black.
- **Morph Operation:** You can choose Shrink or Grow to dilate or erode the edges of the matte, or you can choose Opening or Closing to plug or expand holes to clean up a ragged matte.
- **Morph Radius:** Adjusts how much to shrink, grow, open, or close the key.
- **Denoise:** Provides a distinct way to post-process extracted keys to selectively reduce the noise in a key, getting rid of stray areas of qualification and softly filling holes in a matte.
- **Shadow:** Adjusts key strength based on the darker parts of the original image.
- **Midtone:** Adjusts key strength based on the midtones of the original image.
- **Highlight:** Adjusts key strength based on the brighter parts of the original image.
- **Post Filter:** Performs a final clean-up of the key, using the original image for reference; useful for bringing back some fine detail in sharp edges or hair.

Garbage Matte

These controls let you crop out other unwanted things in the frame, such as lighting fixtures, boom microphones, tracking patches, etc.

- **Matte Shape:** You can choose either a Rectangle or Ellipse with which to crop around the subject you're keying. Choosing None turns off Garbage Matte. When a garbage matte is enabled, the Open FX Overlay mode of the Viewer shows an on-screen control for adjusting the shape and position of the garbage matte.
- **Edge Softness:** Lets you soften the edges. Defaults to 50; the range is 0 (no softening) to 100 (maximum softening).
- **Invert:** Inverts the garbage matte.
- **Center X and Y:** Lets you adjust and keyframe the position of a garbage matte.
- **Bounding Width and Height:** Lets you adjust the width and height of the garbage matte.
- **Rotation:** Lets you adjust the angle of the garbage matte.

Output

The Output controls let you choose how the image is output, composited, and displayed in the Viewer.

- **Output:** This menu offers three options for which channels are output from the effect.
 - Alpha Highlight:** Displays the transparent part of an Alpha channel as gray and the solid part of an alpha channel in full color.
 - Alpha Highlight B/W:** Displays the black and white Alpha channel, which is often helpful when sampling additional areas to key.
 - Final Composite:** The keyed image is displayed composited over any video track below it.
- **Use Alpha:** A common control in many Resolve FX and 3rd party Open FX plugins; this checkbox determines if the Alpha channel is used when compositing over another video track. It can be used instead of choosing RGB Only (Blank Alpha) from the Output menu.

To use the HSL Keyer in the Edit page:

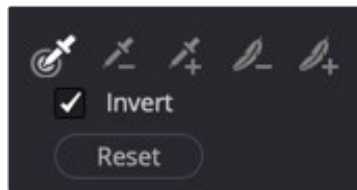
- 1 Apply the HSL Keyer to a clip on video track 2 or higher.
- 2 From the Viewer Overlay menu in the lower-left corner of the Timeline Viewer choose Open FX Overlays.
- 3 Click and drag across the screen color you want to remove.
- 4 The transparency immediately takes effect, showing the keyed subject against whatever clip appears on the video track underneath it in the Timeline as a composite. To see the matte you created for further adjustment, choose Alpha Highlight or Alpha Highlight B/W from the Output drop-down menu.
- 5 To add or subtract from the matte, click the plus or minus Color Range control, and click or drag across the portion of the keyed image.
- 6 To add softness to the outer range of the key you're creating, click the plus Softness control and then click or drag across the portion of the image you'd like to include as a soft edge.

Luma Keyer

Based on the Luma Qualifier in the Color page, the Luma Keyer pulls a key from the luma channel. It's identical to the HSL Keyer with H and S turned off. Although not often used by itself, the Luma Keyer can produce useful effects when used in conjunction with Composite modes.

Selection Range Controls

The Selection Range buttons in the Inspector let you define a key by sampling pixels in the Viewer with the mouse pointer.



Luma Keyer Selection Range controls

- **Sample Eyedropper:** The first control you use whenever sampling the image. It defines the initial range of the keyer; you must use this tool before any of the others. In the Edit page, the Timeline Viewer Overlay must be set to Open FX. Clicking once with this tool selects a single pixel value, while clicking and dragging selects a range of image values that add together.
- **Add/Subtract Luma Range:** These two controls let you add areas of the image to, or subtract them from, the currently selected luma range that defines the core of the key. As with the eyedropper, you can click on single pixels, or drag over a range of color.
- **Add/Subtract Softness:** These two controls let you redefine the softness that transitions from the inner range of the key, falling off towards the outer edge of the key. Like the other keying controls, you can click on single pixels, or drag over a range of color.
- **Invert:** Inverts the sampled luma range to become opaque, and areas not sampled become transparent.
- **Reset:** Resets the sampled luma range while retaining the Output control settings.

Matte Finesse

You may find that sampling the screen color is not enough to overcome some problems. Issues such as chattery edges, holes, or noise can sometimes be fixed using the Matte Finesse controls. These controls filter the output of the Keyer and are adjustments that are made to the matte itself, modifying the key using the frame's content.

- **Pre Filter:** This slider attempts to clean up the image before colors are sampled. This adjustment can be useful when you have footage containing MPEG blocking artifacts.
- **Clean Black:** Clean Black is a specialized operation that eliminates noise seen as white speckling in the black area of a key. Raising Clean Black lets you “fill holes” in the background portion of a key and erode translucent edges.
- **Clean White:** Clean White is another specialized operation that eliminates noise (seen as black speckling when viewing a high-contrast highlight) in the white portion of a key that includes areas of the image you're isolating and expands the key by making light parts of a key lighter the higher you raise this parameter, pushing light gray areas of the key toward white. The practical result is that raising Clean White lets you “fill holes” in the foreground portion of a key and grow translucent edges.

- **Black Clip:** Raising Black Clip applies a “lift” adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed toward black. The range is 0 to 100, with 0 being the default setting.
- **White Clip:** Lowering White Clip applies a “gain” adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed toward white. The range is 0 to 100, with 100 being the default setting.
- **Blur Radius:** In small amounts, blurring a key does well to take the edge off problem edges. However, blurring a key can also feather the edges of a key past the border of the subject you’re keying, with the result being a visible “halo” around your subject, depending on the adjustment you’re making. The range is 0 to 2000, with 0 being the default. With such a large maximum blur radius, combined with the capabilities that the In/Out Ratio provides in customizing the direction of spread, you can turn some pretty precarious mattes into surprisingly smooth and useful results.
- **In/Out Ratio:** Using In/Out Ratio can help eliminate fringing. Raising In/Out Ratio will fill in small black holes in the matte, while lowering In/Out Ratio below 0 will eliminate speckling by pushing small white bits of the matte toward black.
- **Morph Operation:** You can choose Shrink or Grow to dilate or erode the edges of the matte, or you can choose Opening or Closing to plug or expand holes to clean up a ragged matte.
- **Morph Radius:** Adjusts how much to shrink, grow, open, or close the key.
- **Denoise:** Provides a distinct way to post-process extracted keys to selectively reduce the noise in a key, getting rid of stray areas of qualification and softly filling holes in a matte.
- **Shadow:** Adjusts key strength based on the darker parts of the original image.
- **Midtone:** Adjusts key strength based on the midtones of the original image.
- **Highlight:** Adjusts key strength based on the brighter parts of the original image.
- **Post Filter:** Performs a final clean-up of the key, using the original image for reference; useful for bringing back some fine detail in sharp edges or hair.

Garbage Matte

These controls let you crop out other unwanted things in the frame, such as lighting fixtures, boom microphones, tracking patches, etc.

- **Matte Shape:** You can choose either a Rectangle or Ellipse with which to crop around the subject you’re keying. Choosing None turns off Garbage Matte. When a garbage matte is enabled, the Open FX Overlay mode of the Viewer shows an on-screen control for adjusting the shape and position of the garbage matte.
- **Edge Softness:** Lets you soften the edges. Defaults to 50; the range is 0 (no softening) to 100 (maximum softening).
- **Invert:** Inverts the garbage matte.
- **Center X and Y:** Lets you adjust and keyframe the position of a garbage matte.
- **Bounding Width and Height:** Lets you adjust the width and height of the garbage matte.
- **Rotation:** Lets you adjust the angle of the garbage matte.

Output

The Output controls let you choose how the image is output, composited, and displayed in the Viewer.

- **Output:** This menu offers three options for which channels are output from the effect.
 - Alpha Highlight:** Displays the transparent part of an Alpha channel as gray and the solid part of an alpha channel in full color.
 - Alpha Highlight B/W:** Displays the black and white Alpha channel, which is often helpful when sampling additional areas to key.
 - Final Composite:** The keyed image is displayed composited over any video track below it.
- **Use Alpha:** A common control in many Resolve FX and 3rd party Open FX plugins; this checkbox determines if the Alpha channel is used when compositing over another video track. It can be used instead of choosing RGB Only (Blank Alpha) from the Output menu.

To use the Luma Keyer in the Edit page to key a subject:

- 1 Apply the Luma Keyer to a clip on video track 2 or higher.
- 2 From the Viewer Overlay menu in the lower-left corner of the Timeline Viewer choose Open FX Overlays.
- 3 Either click a pixel in a bright or dark part of the image you want to key out, or click and drag across a range of pixels within that subject.
- 4 The transparency immediately takes effect, showing the keyed subject against whatever clip appears on the video track underneath it in the Timeline as a composite. To see the matte you created for further adjustment, choose Alpha Highlight or Alpha Highlight B/W from the Output drop-down menu.
- 5 To add or subtract from the matte, click the plus or minus Luma Range buttons, and then click or drag across the portion of the keyed image.
- 6 To add softness to the outer range of the key you're creating, click the plus Softness button and then click or drag across the portion of the image you'd like to include as a soft edge.